

TRUBCHEVSK, Russian Federation

WMO: 269970

Lat: 52.583N

Lon: 33.767E

Elev: 584

StdP: 14.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-2.3	-13.0	2.8	-8.1	-7.1	3.8	-2.0	22.6	29.8	18.8	26.1	2.9	30	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	86.2	68.6	82.9	67.3	79.8	66.1	71.1	81.1	69.4	79.3	67.8	77.3	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	104.1	76.8	65.9	97.7	74.6	64.4	92.5	72.4	35.3	81.1	33.9	79.4	32.5	77.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.6	15.6	12.6	DB	-13.8	90.6	5.8	3.6	-18.0	93.1	-21.4	95.2	-24.7	97.2	-28.9	99.8
(5)				WB	-14.1	74.3	5.7	2.5	-18.3	76.1	-21.6	77.6	-24.8	79.0	-29.0	80.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	21.5	22.6	31.9	46.6	57.5	63.6	67.2	64.7	54.9	43.8	33.6	25.1
	DBStd	18.46	10.86	11.48	8.40	7.90	8.15	6.42	5.55	6.26	7.13	7.62	8.98	10.73
	HDD50	3915	882	767	563	157	25	1	0	0	32	221	494	773
	HDD65	7740	1347	1187	1027	551	255	103	40	83	310	657	942	1238
	CDD50	1923	0	0	1	56	259	408	534	455	179	29	2	0
	CDD65	273	0	0	0	0	24	60	109	73	7	0	0	0
	CDH74	2541	0	0	0	8	258	507	917	759	91	1	0	0
	CDH80	670	0	0	0	1	45	100	266	249	9	0	0	0
Wind	WSAvg	5.2	5.8	5.8	6.1	5.6	5.2	4.9	4.2	4.3	4.5	5.1	5.6	5.8
Precipitation	PrecAvg	24.2	1.6	1.4	1.3	1.5	2.4	2.6	3.2	2.2	2.3	2.1	1.9	1.6
	PrecMax	31.2	3.1	2.7	3.1	3.8	4.1	6.8	6.8	5.2	4.9	5.0	3.6	3.3
	PrecMin	16.6	0.4	0.6	0.2	0.1	0.7	0.4	0.4	0.0	0.2	0.2	0.0	0.5
	PrecStd	3.3	0.6	0.6	0.6	0.8	0.8	1.2	1.6	1.3	1.1	1.2	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	42.3	58.5	74.1	84.6	86.4	91.1	93.2	81.0	69.5	56.1	47.0
		MCWB	40.0	39.2	47.5	55.3	66.4	68.1	70.0	68.0	63.2	58.2	52.8	45.1
	2%	DB	37.8	38.7	52.0	69.2	80.5	82.9	86.4	86.2	76.7	63.1	50.5	43.0
		MCWB	36.3	36.4	43.3	54.0	64.1	67.2	69.3	68.3	62.6	56.1	48.3	41.5
	5%	DB	35.8	36.8	47.0	64.7	76.6	79.8	82.6	81.7	71.4	58.4	47.3	38.9
		MCWB	34.6	35.0	40.2	52.3	62.6	65.9	68.5	67.1	60.1	53.5	45.4	37.8
	10%	DB	33.7	35.2	42.9	60.4	72.5	76.5	79.1	77.6	67.0	54.7	44.7	36.3
		MCWB	32.7	33.8	38.3	50.5	59.9	64.7	67.6	64.7	58.2	51.1	43.1	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	39.9	48.5	58.5	69.9	71.9	74.1	73.2	66.3	59.8	52.8	45.5
		MCDB	41.2	41.4	57.0	68.9	80.8	81.6	84.9	83.6	77.3	65.4	55.6	46.6
	2%	WB	36.6	37.1	44.1	55.5	65.9	69.3	71.6	70.3	63.4	56.9	48.8	41.5
		MCDB	37.5	38.1	50.4	64.7	76.2	79.2	81.5	81.1	73.2	62.5	50.2	42.7
	5%	WB	34.7	35.5	41.3	53.6	63.3	67.4	70.0	68.2	61.2	54.0	45.8	37.9
		MCDB	35.6	36.4	45.7	62.4	73.7	77.1	79.4	79.7	68.8	57.7	47.0	38.7
	10%	WB	32.9	33.9	38.7	51.7	61.1	65.6	68.5	66.2	59.2	51.5	43.2	35.4
		MCDB	33.7	34.9	42.3	59.8	70.8	74.2	77.1	75.5	65.6	54.3	44.6	36.1
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.8	12.8	17.1	19.3	17.9	18.1	20.0	17.1	12.9	7.9	7.7
		MCDBR	6.4	8.8	18.3	23.8	24.5	22.9	23.8	26.7	24.6	19.2	10.5	7.8
		MCWBR	6.1	7.7	11.0	12.0	11.1	10.6	10.2	11.3	12.3	12.5	8.5	7.2
	5% WB	MCDBR	6.1	8.0	16.7	20.0	22.3	20.3	20.5	23.1	21.4	17.6	9.3	7.5
		MCWBR	6.0	7.7	11.0	11.2	11.2	10.5	10.6	11.2	12.0	12.5	8.1	7.2
Clear-Sky Solar Irradiance	taub	0.299	0.332	0.359	0.428	0.398	0.385	0.420	0.425	0.389	0.357	0.336	0.304	
	taud	2.138	2.069	2.138	2.129	2.303	2.361	2.279	2.248	2.313	2.367	2.259	2.178	
	Ebn at Noon	210	237	259	254	271	275	262	253	248	232	196	185	
	Edh at Noon	23	33	38	44	39	37	39	38	32	25	21	19	
All-Sky Solar Radiation	RadAvg	252	543	986	1306	1763	1858	1765	1529	1011	517	208	153	
	RadStd	49	82	148	120	130	163	133	149	113	92	39	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.13	N/A	N/A	+1.69	N/A	N/A	-256	-423	+185	N/A
(47)	Regional (9 neighbors)	+1.22	N/A	N/A	+1.62	N/A	N/A	-252	-382	+189	+74

Nomenclature: See separate page